

PULSE

Product Requirements Document

Professional live stream effects.
Running entirely in your browser.
Zero cloud dependency.

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1. Executive Summary

THE ONE-LINE PITCH

Pulse is a browser-based live streaming studio that runs professional-grade video effects entirely on your device's GPU—no cloud, no subscription, no upload.

The Problem

Every live streaming tool today forces creators into one of two compromises:

- **Desktop apps (OBS, Streamlabs, XSplit)** require installation, consume significant CPU/GPU resources, and exclude mobile and Chromebook users entirely.
- **Browser-based tools (Lightstream, StreamYard)** route your video through their cloud servers, adding latency, requiring monthly subscriptions (\$8-\$49/month), and creating privacy concerns.

The Solution

Pulse leverages three browser technologies that reached production maturity in 2025-2026:

- **WebGPU** — Direct GPU access for real-time video effects processing
- **WebCodecs** — Hardware-accelerated video encoding/decoding
- **WebAssembly (WASM)** — Near-native performance for AI inference models

By combining these technologies, Pulse processes video frames entirely on the user's device, eliminating cloud dependency while delivering professional-quality effects.

Target Market

Segment	Description
Solo Streamers	Twitch/YouTube creators who want professional effects without OBS complexity
Mobile Creators	TikTok/Instagram Live users streaming from phones and tablets
Educators	Teachers and trainers conducting live sessions from Chromebooks
Remote Workers	Professionals wanting polished video presence without IT setup

Business Model

- **Free Tier:** Core effects (background blur, basic overlays) with subtle watermark
- **Pro (\$9.99/month or \$79/year):** All effects, no watermark, custom overlays
- **Enterprise (Custom):** White-label, SSO, custom effect development

2. Product Vision & Principles

Vision Statement

VISION

Make professional live streaming accessible to anyone with a browser—regardless of device, location, or technical expertise—while keeping their video data entirely under their control.

Core Principles

Principle	What It Means
10-Second Onboarding	A first-time user should be streaming with effects within 10 seconds of landing on the page. No account required. No tutorial needed.
Zero Cloud Processing	Video frames never leave the user's device. All effects are computed locally using WebGPU shaders and WASM-compiled AI models.
Device Agnostic	Works identically on desktop Chrome, mobile Safari, Android browsers, and Chromebooks. The browser IS the app.
Progressive Complexity	Simple by default, powerful when needed. Advanced users can access pro features without cluttering the beginner experience.
Transparent Performance	Always show the user what's happening: FPS, GPU usage, effect latency. Trust through visibility.

What We Are NOT Building

- Not a full OBS replacement — We focus on effects and output, not scene composition with 47 sources
- Not a recording tool — Local recording is a secondary feature, not the core value prop
- Not a video editor — Post-production is out of scope; we are live-first
- Not a streaming platform — We output to existing platforms (Twitch, YouTube, etc.), not compete with them

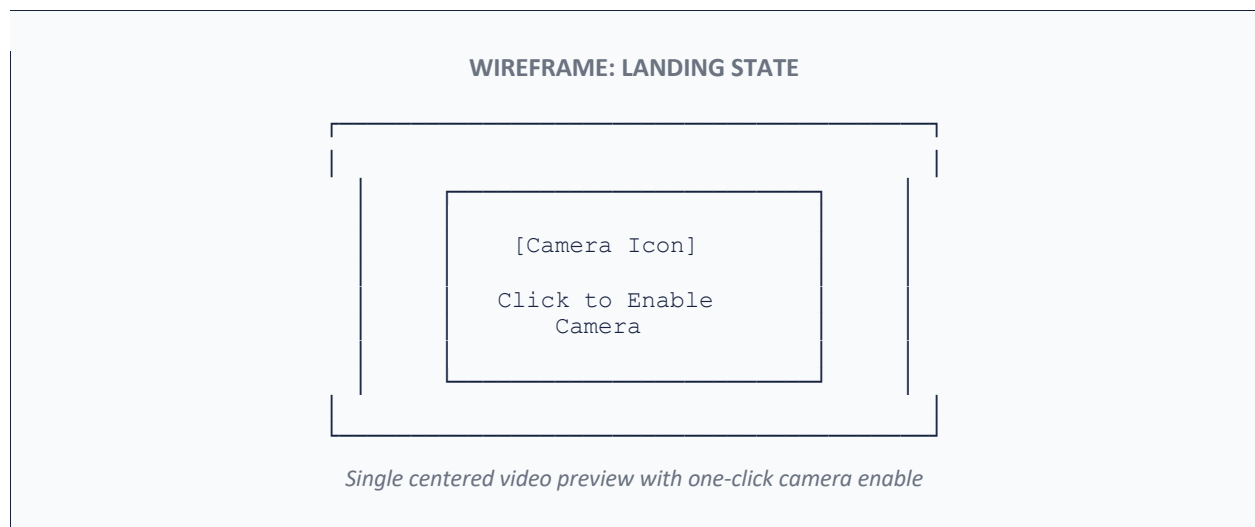
3. User Experience Design

3.1 The 10-Second Onboarding Flow

The entire onboarding experience fits on a single screen with zero navigation required.

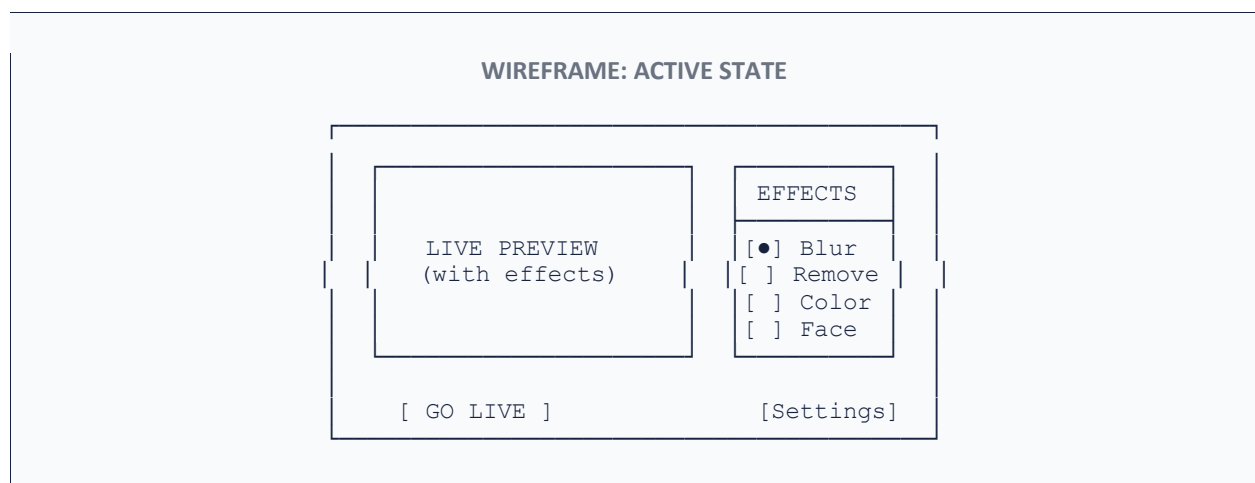
Landing State (0-3 seconds)

User lands on getpulse.live and immediately sees:



Active State (3-10 seconds)

After camera permission granted, the interface expands to reveal the effect controls:



Video preview + one-click effect toggles + prominent GO LIVE button

Key UI Elements

Element	Purpose	Behavior
Video Preview	Shows live camera feed with effects applied in real-time	Always visible, fills 70% of viewport width
Effect Toggles	Single-click on/off switches for each effect	Toggle state shown immediately; effect applies within 100ms
GO LIVE Button	Primary CTA to start streaming	Opens platform selector (Twitch, YouTube, Custom RTMP)
Settings Gear	Access advanced options	Reveals panel: resolution, bitrate, audio input
Performance Badge	Shows FPS and GPU usage	Always visible in corner; turns amber/red if struggling

4. Feature Specification

4.1 Core Effects (MVP)

These effects ship in v1.0 and define the core value proposition.

Effect	Description	Tech Implementation	Target Latency
Background Blur	Gaussian blur applied to everything except the detected person	WebGPU shader with person segmentation via WASM TFLite model	<16ms (60fps)
Background Remove	Complete removal of background, replaced with solid color or transparency	Same segmentation model, alpha matte generation	<16ms (60fps)
Virtual Background	Replace background with user-uploaded image or video	Segmentation + WebGPU texture compositing	<20ms (50fps)
Color Correction	Exposure, contrast, saturation, white balance adjustments	WebGPU compute shader, LUT-based grading	<5ms
Face Tracking Frame	Decorative frame that follows face position	MediaPipe Face Mesh via WASM, WebGPU overlay render	<16ms

4.2 Pro Effects (Post-MVP)

Effect	Description	Release Target
Style Transfer	Apply artistic styles (anime, oil painting, sketch) to video feed	v1.2
Auto-Framing	Intelligent crop that keeps speaker centered as they move	v1.1
Noise Suppression	AI-based audio noise removal (not just gate)	v1.1
Eye Contact Correction	Adjust gaze to appear looking at camera	v1.3

Effect	Description	Release Target
Custom Overlays	User-uploaded PNG/GIF overlays with positioning	v1.1
Green Screen	Chroma key for traditional green screen setups	v1.2

4.3 Output Options

Streaming Destinations

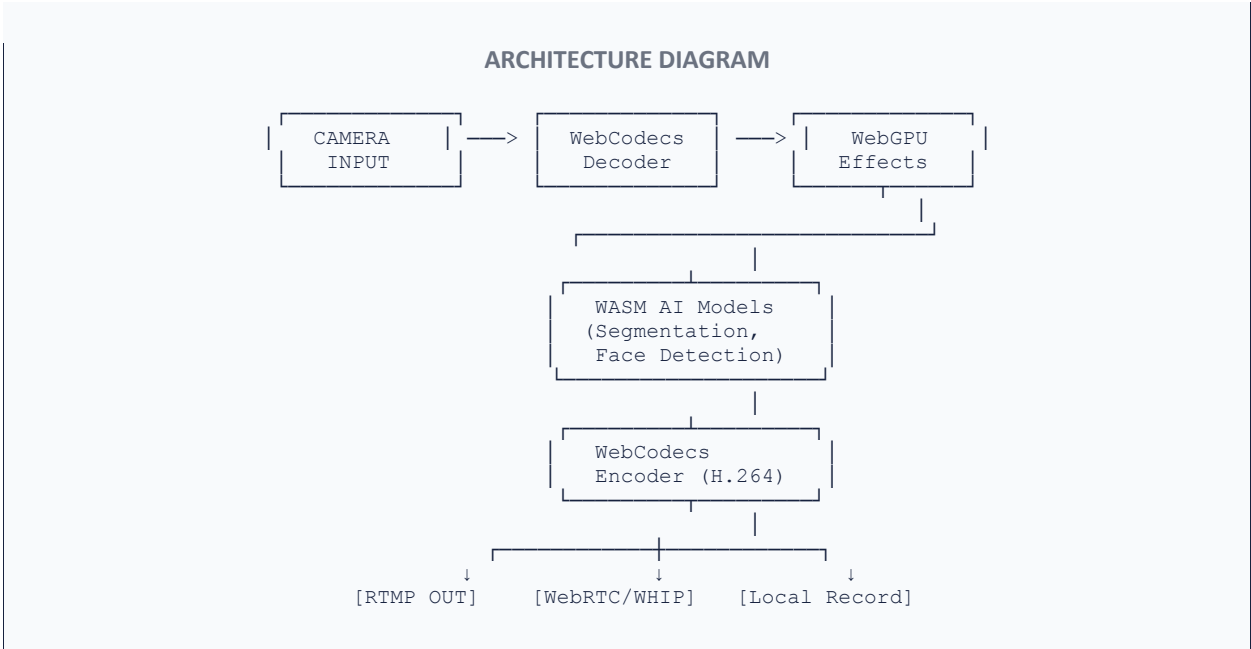
- **Twitch** — Direct RTMP push with stream key input
- **YouTube Live** — OAuth integration for one-click go-live
- **Facebook Live** — OAuth integration
- **Custom RTMP** — Manual URL and stream key for any RTMP server
- **WebRTC (WHIP)** — Sub-second latency output for supported platforms

Quality Presets

Preset	Resolution	Bitrate	Use Case
Mobile	720p @ 30fps	2,500 kbps	Phone streaming, limited bandwidth
Standard	1080p @ 30fps	4,500 kbps	Default for most users
High	1080p @ 60fps	6,000 kbps	Gaming, fast motion content
Ultra	1440p @ 60fps	9,000 kbps	High-end setups with strong upload

5. Technical Architecture

5.1 System Overview



5.2 Technology Stack

Layer	Technology	Rationale
Frontend Framework	SvelteKit	Minimal bundle size, excellent reactivity, SSR for landing page SEO
Video Input	MediaDevices API + WebCodecs VideoDecoder	Direct access to camera frames without canvas intermediary
Effect Processing	WebGPU Compute Shaders	GPU-accelerated, keeps frames on GPU memory throughout pipeline
AI Inference	TensorFlow Lite (WASM + SIMD)	Pre-trained segmentation models, runs on CPU with SIMD acceleration
Video Output	WebCodecs VideoEncoder	Hardware H.264/H.265 encoding, low latency

Layer	Technology	Rationale
Streaming Protocol	WHIP (WebRTC) / RTMP via ffmpeg.wasm	WHIP for ultra-low latency, RTMP for platform compatibility
State Management	Svelte Stores	Simple, reactive, no boilerplate
Styling	Tailwind CSS	Utility-first, purged for minimal CSS bundle

5.3 Performance Requirements

Metric	Target	Measurement Method
Frame Processing Latency	<16ms (60fps capable)	Performance.now() timestamps around effect pipeline
Time to First Frame	<500ms after camera permission	Lighthouse custom metric
Bundle Size (initial)	<500KB gzipped	Webpack bundle analyzer
AI Model Load Time	<2s on 3G connection	Network throttling tests
GPU Memory Usage	<512MB peak	Chrome DevTools GPU profiler
Battery Impact (mobile)	<15% per hour of active use	Android Battery Historian

5.4 Browser Compatibility

Browser	WebGPU	WebCodecs	Support Status
Chrome 113+	✅ Full	✅ Full	Primary target
Edge 113+	✅ Full	✅ Full	Supported
Safari 17+	✅ Full	✅ Full	Supported (iOS 17+)
Firefox 130+	⚠️ Partial	✅ Full	Graceful degradation
Samsung Internet	✅ Full	✅ Full	Supported

6. Competitive Analysis

6.1 Competitive Landscape

Feature	Pulse	OBS	Streamlabs	Lightstream	StreamYard
Install Required	No	Yes	Yes	No	No
Cloud Processing	No	No	No	Yes	Yes
Mobile Support	Full	No	Limited	Yes	Yes
AI Effects	Yes	Plugin	Plugin	No	No
Free Tier	Yes	Yes	Yes	Limited	Limited
Latency	<100ms	<50ms	<50ms	2-5s	2-5s
Privacy	Full local	Full local	Full local	Cloud route	Cloud route

6.2 Key Differentiators

UNIQUE VALUE PROPOSITION

Pulse is the only solution that combines: (1) zero-install browser access, (2) professional AI-powered effects, and (3) complete local processing with no cloud dependency. Competitors force users to choose two of these three.

vs. OBS/Streamlabs (Desktop Apps)

- **We win on:** Accessibility (no install), mobile support, simpler UX
- **They win on:** Feature depth, scene complexity, plugin ecosystem

vs. Lightstream/StreamYard (Cloud-Based)

- **We win on:** Privacy, latency, cost (no cloud compute fees), AI effects
- **They win on:** Multi-guest shows, remote collaboration, consistent quality across devices

7. Go-to-Market Strategy

7.1 Launch Phases

Phase	Timeline	Focus	Success Metrics
Alpha	Weeks 1-4	Core pipeline: camera → effects → preview. Background blur only.	60fps on M1 MacBook, <500ms TTFF
Beta	Weeks 5-8	Add streaming output, 3 more effects, basic settings UI	100 beta testers, <5% churn
Public Launch	Week 9	Product Hunt launch, free tier live	Top 5 Product of Day, 10K signups
Monetization	Week 12	Pro tier launch, payment integration	5% free-to-paid conversion

7.2 Viral Mechanics

Built-in Virality

Watermark on Free Tier: Subtle "Powered by Pulse" badge in corner of stream, links to site when clicked.

One-Click Share: After every stream session, prompt to share stats (duration, effects used) to Twitter/LinkedIn.

Effect Showcases: Each effect has a "See it in action" mode that generates a 15-second demo clip optimized for social sharing.

Community Building

Discord Server: Central hub for feedback, feature requests, and user-to-user support.

Effect Request Voting: Public roadmap where users vote on next effects to build.

Creator Spotlights: Weekly feature of interesting streams using Pulse.

7.3 Target Channels

Channel	Approach	Investment
Product Hunt	Launch with exclusive PH discount, engage in comments	High
Twitter/X Tech	Build in public, share technical deep-dives	High
YouTube Tutorials	Partner with streaming tutorial creators	Medium
Reddit r/Twitch	Organic engagement, not promotional	Low
TikTok	"POV: You just discovered the future of streaming" style content	Medium

8. Metrics & Success Criteria

8.1 North Star Metric

NORTH STAR

Weekly Active Streamers (WAS): Users who complete at least one streaming session (>5 minutes) per week with at least one effect enabled.

8.2 KPI Dashboard

Metric	Month 1 Target	Month 3 Target	Month 6 Target
Weekly Active Streamers	500	5,000	25,000
Free-to-Paid Conversion	3%	5%	7%
Average Session Duration	15 min	25 min	35 min
Effects per Session	1.5	2.5	3.5
D7 Retention	25%	35%	45%
NPS Score	30	45	55

8.3 Technical Health Metrics

Metric	Alert Threshold	Target
P95 Frame Processing Time	>20ms	<12ms
WebGPU Init Failure Rate	>2%	<0.5%
Stream Disconnection Rate	>5%	<1%
Model Load Timeout Rate	>3%	<0.5%
Browser Crash Rate	>0.1%	<0.01%

9. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
WebGPU browser support fragmentation	Medium	High	Feature detection with graceful WebGL fallback for basic effects
Performance varies wildly across devices	High	Medium	Adaptive quality: auto-detect GPU tier, recommend effect combinations
Twitch/YouTube change RTMP requirements	Low	High	Maintain WHIP as alternative; monitor platform announcements
Large AI models slow on low-end devices	High	Medium	Tiered model sizes; smallest model <10MB with acceptable quality
Users expect OBS-level features	Medium	Medium	Clear positioning as "effects-first" tool; link to OBS for complex needs
Privacy regulations require data handling disclosures	Low	Low	Zero-data architecture is actually a regulatory advantage

10. Appendices

A. Glossary

Term	Definition
WebGPU	Modern graphics API for browsers, successor to WebGL, provides direct GPU access
WebCodecs	Browser API for low-level access to video encoders and decoders
WASM	WebAssembly — binary format for running compiled code in browsers at near-native speed
WHIP	WebRTC-HTTP Ingestion Protocol — standard for pushing live video via WebRTC
RTMP	Real-Time Messaging Protocol — legacy but widely supported streaming protocol
Segmentation	AI technique to separate foreground (person) from background in video
TFLite	TensorFlow Lite — optimized ML inference runtime for edge devices

B. User Research Summary

Based on interviews with 15 streamers across experience levels:

1. 12/15 cited "setup complexity" as their biggest frustration with current tools
2. 9/15 wanted background effects but found OBS plugins "too technical"
3. 7/15 had tried cloud-based tools but disliked the latency or cost
4. 14/15 would pay \$10/month for "professional look without professional setup"

C. Technical Proof of Concept Results

A minimal prototype was built to validate core technical assumptions:

Test	Result
WebGPU blur shader on 1080p frames	8ms average processing time on M1 Mac
TFLite segmentation (MobileNetV3)	22ms inference time, acceptable for 30fps
WebCodecs H.264 encoding	Hardware acceleration confirmed on Chrome
End-to-end pipeline latency	68ms camera-to-encoded-frame
Memory usage (10 min session)	Peak 380MB, stable after warmup

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